MILITARY SPECIFICATION

RELAY, ARMATURE

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for two types of armature relays entitled as Relay, Sensitive and Relay, Control used in the Honest John warhead section.

1.2 Classification. This specification covers relays of the following:

Type I Relay, Sensitive.
Type II Relay, Control.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8806618	- Bag, Packing, Ammunition for Spare Parts and Adaption Kit Parts.
8808362	- Container, Packing, Ammunition for Domestic (CONUS) Shipment of Special Weapons Items, Packing and Marking.
8813566	- Relay, Control.
8831126	- Relay, Sensitive.
EL-8842548	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The relays shall comply with all requirements specified on the drawing (dwg) 8813566 and 8831126, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial code shall be assigned to the item. Serial codes required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial codes required (unused code numbers shall be returned).

3.5 Coil resistance.

3.5.1 Sensitive relay. The resistance between pin 1 and pin 3 of coil A shall be 0.38 plus or minus (±) 0.04 ohms and the resistance between pin 4 and pin 7 of the coil B shall be 20,000 ± 2,000 ohms when determined as specified in 4.4.1.

3.5.2 Control relay. The coil resistance shall be between 300 and 500 ohms when determined as specified in 4.4.1.

3.6 Pull-in current, sensitive relay. The pull-in current of coil A shall be 30 ± 12 milliamperes (ma) direct current (dc) when measured within the temperature range of -65 ± 5 degrees Fahrenheit (°F) to 160 ± 5°F. The pull in current of coil B shall be 0.170 ± 0.051 milliamperes dc. The the temperatures of -65 ± 5°F and 160 ± 5°F, the pull-in current of coil B shall be within 30 percent of the value obtained at 80 ± 20°F. The dropout to pull-in current differential shall be 3 milliamperes maximum on coil A and 0.02 milliamperes maximum on coil B. Testing shall be as specified in 4.4.2.

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3.7 Pull-in voltage, control relay. The pull-in voltage shall be a maximum of 20 volts (v) dc. The time required for the normally open relay contacts to close shall be a maximum of 30 milliseconds from the time 24 vdc is applied to the coil. Testing shall be as specified in 4.4.3.

3.8 Dielectric strength. The sensitive relay shall have no indication of dielectric breakdown with 600 ± 60 volt root mean square (vrms), 60 Hertz (Hz), applied between the case and each of the following pins: 1,2,4,6, and 8 and between the following pairs of pins: 1 to 2, 1 to 4, 1 to 6, 1 to 8, 2 to 4, 4 to 6, and 4 to 8. The control relay shall have no indication of dielectric breakdown with 500 ± 60 vrms, 60 Hz, applied between the case and each of the following terminals 1,3,4,6, and 8 and between the following pairs of pins: 1 to 2, 1 to 4, 1 to 6, 1 to 8, 2 to 4, 2 to 6, 2 to 8, 4 to 6, 4 to 8, and 6 to 8. Testing shall be as specified in 4.4.4.

3.9 Contact resistance, sensitive relay. With an input voltage of 110 percent of the voltage just sufficient to pull in the contacts, the contact resistance measured from pins 2 to 8 and 6 to 8 shall be less than 3 ohms when checked at 80 ± 20°F, -65 ± 5°F, and 160 ± 5°F. Testing shall be as specified in 4.4.5.

3.10 Repeatability, sensitive relay. While at a stabilized ambient temperature (see 6.4), the relay shall not operate when a sharply rising current, having a value of 90 percent of the actual measured pull-in current, is applied to the coil. After a pulse of 4 times the pull-in current is applied to the coil, the relay shall still fail to operate with the above reduced current. Testing shall be as specified in 4.4.6.

3.11 Environmental testing. The relays shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The relays shall also comply with 3.5 through 3.10 as applicable before and after each environmental test. Testing shall be as specified in 4.4.11. Tests shall be conducted in the order specified.

3.11.1 High temperature. The relays shall withstand a temperature of 160 ± 5°F without swelling, peeling, cracking or other damage and the relays shall comply with 3.5 through 3.10 as applicable, while at 160 ± 5°F when tested as specified in 4.4.7.1.

3.11.2 Low temperature. The relays shall withstand a temperature of 80 ± 5°F without cracking, peeling, swelling or other damage and the relays shall comply with 3.5 through 3.10 as applicable, while at -65 ± 5°F when tested as specified in 4.4.7.2.

3.11.3 Temperature shock. The relays shall withstand exposure to temperature shocks of 160 ± 5°F to 80 ± 20°F to -65 ± 5°F without cracking, peeling, swelling or other damage when tested as specified in 4.4.7.3. Upon completion of this test the relays shall comply with 3.5 through 3.10 as applicable.

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3.11.4 Vibration. The relays shall withstand vibration without cracking, splitting or other damage, in each of their three major and mutually perpendicular axes at the frequencies, gravity units (g's), amplitudes, temperatures and times specified in Table I.

Table I

Vibration conditions

Assembly	Test Cycle Frequency Hz	Double Amplitude (Inches)	g Level	Total Cycling Time (Minutes) 80 ± 20°F	160 ± 5°F	-65 ± 5°F
sensitive relay	5 to 75	±0.030±0.006		15 min*	15 min	15 min
	75 to 500		10 ± 1	15 min	15 min	15 min
control relay	10 to 500		10 ± 1	15 min	15 min	15 min

*Minimum

In addition to the time speicifed in Table I, the relays shall withstand vibration for three minutes at the most severe resonant node encountered. The amplitude of this resonant frequency shall be in accordance with Table I. Testing shall be as specified in 4.4.7.4.

3.11.5 Shock. The relays shall withstand six shocks of the type specified in Table II in each of their three major mutually perpendicular axes without cracking, splitting or other damage when tested as specified in 4.4.7.5. Upon completion of this test the relays shall comply with 3.5 through 3.10, as applicable.

Table II

Assembly	Peak Shock Amplitude	Time Duration Half Amplitude (Milliseconds)
Sensitive relay	20 g	10 ± 1
Control relay	15 g	10 ± 1

3.11.6 Leakage. The leakage rate shall not exceed 1×10^{-5} cubic centimeter per second when tested as specified in 4.4.7.6.

3.12 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.13 Workmanship.

3.13.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.13.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.13.3 Burrs. Parts shall be free from burrs.

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3.13.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.13.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.13.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 5 complete assemblies of each type (see dwg 8813566 and 8831126) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table III. Tests shall be conducted in order specified in Table III.

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Table III

First article inspection and tests

Inspection or Test	Relay Type Sensitive	Control	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X		X
2. Coil resistance (see 3.5)	X	X		X
3. Pull-in current (see 3.6)	X			X
4. Pull-in voltage (see 3.7)	X	X		X
5. Dielectric strength (see 3.8)	X	X		X
6. Contact resistance (see 3.9)	X			X
7. Repeatability (see 3.10)	X			X
8. High temperature (see 3.11.1)	X	X	X	
9. Low temperature (see 3.11.2)	X	X		X
10. Temperature shock (see 3.11.3)	X	X		X
11. Vibration (see 3.11.4)	X	X		X
12. Shock (see 3.11.5)	X	X		X
13. Leakage (see 3.11.6)	X	X		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table IV and Table V), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table IV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table V
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Sensitive relay and control relay, prior to sealing (see dwgs 8831126 and 8813566).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Internal leads cut, nicked or damaged	Visual
	102. Improper soldering (see 3.12)	Visual
	103. Evidence of poor workmanship (see 3.13)	Visual
	104. Any component missing, loose or damaged	Visual/Manual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Sensitive relay and control relay (see dwgs 8831126 and 8813566).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Improper location of size of mounting studs	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:

201. Marking missing, incomplete, or incorrect.....Visual
202. Serialization improper or missing (see 3.4).....Visual

4.3.2.1.3 Bag, packing, ammunition for spare parts and adaption kit parts (see dwg 8806618).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Wrong size bag	Visual
	102. Bag not sealed	Visual
	103. Air in bag not minimized before sealing	Visual
	104. Marking missing, mislocated or incorrect	Visual
	105. Evidence of poor workmanship (see 3.13)	Visual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.4 Container, packing, ammunition for domestic (CONUS) shipment of special weapons items, packing and marking (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect or insufficient cushioning material	Visual
	102. Marking improper, illegible, smeared, missing, or in wrong location	Visual
	103. Evidence of poor workmanship (see 3.13)	Visual
	104. Packing material visibly wet	Visual
	105. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

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4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Coil resistance	Major
*b.	Pull-in current	Major
**c.	Pull-in voltage	Major
d.	Dielectric strength	Major
*e.	Contact resistance	Major
*f.	Repeatability	Major
g.	High temperature	Major
h.	Low temperature	Major
i.	Vibration	Major
j.	Leakage	Major

*These tests are applicable to the sensitive relay only.
**This test is applicable to the control relay only.

Of these tests, (g) is considered destructive and samples to tested may be returned to the lot.

4.3.3.1 Coil resistance, dielectric strength and leakage. Each relay in the lot shall be tested in accordance with 4.4.1, 4.4.4 and 4.4.7.6. If any item fails to comply with 3.5, 3.8 and 3.11.6, the item shall be rejected and removed from the lot.

4.3.3.2 Pull-in current, contact resistance and repeatability. Each sensitive relay in the lot shall be tested in accordance with 4.4.2, 4.4.5 and 4.4.6. If any item fails to comply with 3.6, 3.9 and 3.10, the item shall be rejected and removed from the lot.

4.3.3.3 Pull-in voltage. Each control relay in the lot shall be tested in accordance with 4.4.3. If any item fails to comply with 3.7, the item shall be rejected and removed from the lot.

4.3.3.4 High temperature, low temperature and vibration. A sample shall be selected at random in accordance with Table VI. The sample shall be tested in accordance with 4.4.7.1, 4.4.7.2 and 4.4.7.4. If any item fails to comply with 3.11.1, 3.11.2 and 3.11.4, the lot shall be rejected.

Table VI

Lot Size	Sample Size
2-8	1
9-180	2
181-500	3
501-800	5
801-1300	7

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8842548). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used. Unless otherwise specified, testing shall be performed at a temperature of 80 ± 20°F.

4.4.1 Coil resistance. The resistance of the coils shall be determined by the use of a suitable electrical resistance measuring instrument connected between the specified terminals. During resistance measurement the voltage applied to coil A of the sensitive relay shall be limited to a maximum of 15 millivolts.

4.4.2 Pull-in current, sensitive relay. A suitable current measuring instrument capable of providing a reading in milliamperes shall be used to measure the specified pull in currents of coils A and B. This current shall be the minimum sharply rising input current that will consistently operate the relay after the coil has been given a current 4 times the nominal current value. Nominal pull-in current is 30 milliamperes for coil A and 0.17 milliamperes for coil B. The over-current shall have the same polarity as the pull-in current. The instrument shall also be used to measure the specified drop-out to pull-in current differential. The differential shall be determined with a contact current of 50 milliamperes dc, pin 8 negative and a load voltage of 25 ± 3 vdc. The drop-out current shall be the maximum current that will allow the relay to drop out consistently when the coil current is dropped sharply from 110 percent of the nominal pull in current (three consecutive drop-outs shall be considered consistent performance).

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4.4.3 Pull-in voltage, sensitive relay. A suitable dc voltage measuring instrument shall be connected to the coil terminals and shall be used to measure the pull-in voltage. Pull-in voltage is the minimum dc voltage which will consistently operate the relay. Suitable instrumentation shall be used to monitor the contact closure time.

4.4.4 Dielectric strength. The dielectric strength between the specified terminals shall be determined with the use of a suitable dielectric test instrument capable of applying the specified voltage. The test instrument shall be current limited to 5 milliamperes to protect human life.

4.4.5 Contact resistance. Suitable voltage and current measuring instruments shall be used to determine the contact resistance at the specified temperature by the voltmeter-ammeter method. The contact resistance shall be determined with a contact current of 50 milliamperes dc, pin 8 negative, and a lead voltage of 25 ± 3 vdc.

4.4.6 Repeatability. Suitable instrumentation capable of applying 4 times the nominal pull-in currents (see 4.4.2) to the coils and capable of detecting contact closing speeds of 30 milliseconds shall be used to check relay repeatability. A sharply rising current of 90 percent of the measured pull-in current (see 4.4.2) shall be applied to coil B. A pulse of 0.68 milliamperes will then be applied to the coil followed by a sharply rising current valued at 90 percent of the measured pull-in current. The procedure shall be repeated for coil A with the exception that the value of the pulse shall be 120 milliamperes.

4.4.7 Environmental testing (see 3.11).

4.4.7.1 High temperature. The relay shall be placed in a test chamber having an internal temperature of 160 ± 5°F. The relay shall be subjected to this temperature for a minimum of 4 hours. After completing this exposure period and while still at this temperature, the relay shall be tested as specified in 4.4.1 through 4.4.6 as applicable.

4.4.7.2 Low temperature. The relay shall be placed in a test chamber having an internal temperature of -80 ± 5°F. The relay shall be subjected to this temperature for a minimum of 4 hours, after which the temperature shall be raised to -65 ± 5°F. It shall be maintained at this level for an additional 1 hour period or until stabilization (see 6.4) is reached, whichever is longer. After completing this exposure period and while still at this temperature, the relay shall be tested as specified in 4.4.1 through 4.4.6 as applicable.

4.4.7.3 Temperature shock. The relay shall be placed within a test chamber having an internal temperature of 160 ± 5°F. The relay shall be subjected to this temperature for a period of 4 hours minimum, at the conclusion of which, and with 5 minutes, it shall be transferred to a chamber having an internal temperature of 80 ± 20°F. The relay shall be

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subjected to this temperature for a period of 4 hours minimum, after which and within 5 minutes the relay shall be transferred to a chamber having an internal temperature of 80 ± 20°F. The relay shall be subjected to this temperature for a period of 4 hours minimum, after which and within 5 minutes the relay shall be transferred to a chamber having an internal temperature of -65 ± 5°F. The assembly shall be subjected to this temperature for a minimum of 4 hours; within 5 minutes, the relay shall be transferred to a chamber having an internal temperature of 80 ± 20°F. It shall be subjected to this temperature for a minimum of 4 hours. This completes 1 cycle. The number of cycles shall be three. At the completion of the third cycle the assembly shall be removed from the test chamber and tested as specified in 4.4.1 through 4.4.6 as applicable.

4.4.7.4 Vibration. The relays shall be magnetically shielded during this test if the vibration machine does not have a degaussing coil. The relay shall be rigidly mounted to a fixture capable of allowing the relay to be vibrated in each of its three major and mutually perpendicular axes. The fixture shall be securely fastened to a suitable vibration producing machine and the relays shall be vibrated at the frequencies, amplitudes, temperatures, and for the times specified in Table I. After the relay has been vibrated in accordance with Table I, a search for resonance shall be made. Resonant frequencies shall be determined in each of the three major and mutually perpendicular axes by varying the frequency of applied vibration slowly through the frequency range specified in Table I at the specified amplitudes or g levels. The relay shall be vibrated in each of the three major and mutually perpendicular axes at 80 ± 20°F for 3 minutes at the most severe resonant node encountered. If no resonance is found, the relay shall be vibrated for a total of 9 minutes (3 minutes in each axis) at the frequency of 55 Hz and an amplitude of 0.030 ± 0.006 inches. After vibration, the relay shall be tested as specified in 4.4.1 through 4.4.6 as applicable.

4.4.7.5 Shock. The relay shall be mounted to a test fixture in a manner simulating the mounting of the relay in the storage tester. The fixture shall then be mounted to a suitable shock producing machine and the relay subjected to the specified number of shocks in the specified directions.

4.4.7.6 Leakage. The sealed relay shall be tested by utilizing a leak detection method involving the passage of tracer gas from one side of the presumed leak to the other side. The instrument shall provide a method of determining the leakage rate.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The relays shall be packaged in accordance with dwg 8806618 using the specified moisture barrier.

5.1.2 Level C. The relays shall be packaged in accordance with dwg 8806618 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The relays shall be packed in accordance with dwg 8808362.

5.3 Marking. Marking shall be as specified in accordance with dwgs 8806618 and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1689 dated 6 November 1961.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A012

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